

Lecture 6

In Class Exercises



Exercise 1

- Give a brief argument that the running time of PARTITION on a subarray of size n is $\Theta(n)$.
- Modify QUICKSORT to sort into monotonically decreasing order.

Exercise 2

- Use the substitution method to prove that the recurrence $T(n) = T(n - 1) + \Theta(n)$ has the solution $T(n) = \Theta(n^2)$.

Exercise 3

- What value of q does PARTITION return when all elements in the subarray $A[p:r]$ have the same value?
- Modify PARTITION so that $q = (p+r)/2$ when all elements in the subarray $A[p:r]$ have the same value.

PARTITION(A, p, r)

```
1   $x = A[r]$  // the pivot
2   $i = p - 1$  // highest index into the low side
3  for  $j = p$  to  $r - 1$  // process each element other than the pivot
4      if  $A[j] \leq x$  // does this element belong on the low side?
5           $i = i + 1$  // index of a new slot in the low side
6          exchange  $A[i]$  with  $A[j]$  // put this element there
7  exchange  $A[i + 1]$  with  $A[r]$  // pivot goes just to the right of the low side
8  return  $i + 1$  // new index of the pivot
```

Exercise 4

- What is the running time of Quicksort when all elements of array A have the same value?
- Write a recurrence equation for this case.

Exercise 5

- When RANDOMIZED-Quicksort runs, how many calls are made to the random-number generator RANDOM in the worst case? How about in the best case? Give your answer in terms of Θ -notation.